

Path 1: Pre Security > Module 5: Operating Systems Basics > Room 3: Linux CLI Basics

<https://tryhackme.com/room/linuxclibasics>

Linux CLI Basics

Get comfortable navigating through the Linux CLI.

>> Task 1: Introduction

- ◆ Learning Objectives:

- Understand what the Linux terminal is and what it's used for.
- Feel comfortable interacting with the Linux environment.
- Navigate through the Linux filesystem with basic commands.

Q1) What does "CLI" stand for?

Answer: Command-Line Interface

>> Task 2: Navigation Mission: "Find the Missing Notes"

♦ A Quick Note About the Terminal:

Before we begin exploring, let's take a moment to familiarize ourselves with the tool we're using.

The **terminal** is a text-based interface for controlling a Linux system. Instead of interacting with the graphical interface, you type commands that tell the computer exactly what to do. Cyber security professionals use it because:

- It's faster than clicking around
- It gives more control
- Many security tools only run in the terminal

♦ Interacting With the Terminal:

Step 1: "Where Am I?"

When you first open a terminal, you might not know what part of the system you're in.

Use this command to find out: **pwd**

It stands for "print working directory", which basically means "show me the folder I'm currently in".

Step 2: "What's Around Me?"

Now that you know where you are, let's see what files and folders are here: **ls**

This lists the content of the current directory. If we need more details, we can try: **ls -l**

The output displays important information about the files and directories like file sizes, permissions, dates, and more.

Hidden Files:

In order to get the hidden files in the directory, we can append the command to **ls -al**, and it will display all the hidden files present in the directory.

Hidden files aren't really **secret**; they start with a dot **.**, and Linux hides such files by default.

Step 3: Let's Move Around

To walk through the filesystem, we can use: `cd <directory>`. For example: `cd Documents`, and this will change our directory to **Documents**.

To go "back" one level, we will use the command `cd ..`

Step 4: Learn the Power of "Find"

Let's now use one of the handy tools to `find`. As the name suggests, this built-in utility is used to locate files within the file system. Here's a simple version of the command: `find <starting_point> -name <filename>`. Since your supervisor mentioned that the file `mission_brief.txt` resides somewhere in your home directory, begin the home directory symbol: `~`. So we will run the command: `find ~ -name mission_brief.txt`

Step 5: Read the File

Another useful utility is `cat`. This is used to read the content of the file. To read this file, we will run the command `cat mission_brief.txt` to get the content

Q1) What is the full path of the `mission_brief.txt` file found on the system using the `find` command?

Answer: `/home/ubuntu/Documents/.research/archive/mission_brief.txt`

Command: `find ~ -name mission_brief.txt`

Q2) What is the flag hidden inside the `mission_brief.txt` file?

Answer: MISSION-FOUND

Command: `cat mission_brief.txt`

>> Task 3: Investigating the System

Step 1: "Who Are You Logged in As?"

The simplest command in Linux also happens to be one of the most useful: **whoami**

This prints your current username.

Step 2: "What System Are You On?"

To see details about the operating system, kernel version, and architecture, use:
uname -a

This gives a full line of system information - great for understanding exactly what environment you're working in.

Breakdown of the Information

Linux: The system is running the Linux kernel.

tryhackme: The hostname (the computer's name).

<REDACTED>-aws: The kernel version installed on the machine.

x86_64: The hardware platform (also 64-bit).

GNU/Linux: The operating system type (Linux kernel + GNU tools).

If you only want the operating system name, you can try: **uname**

Step 3: Check Disk and Storage Info

In real-world environments, you'll often need to check disk usage or available space, especially before running tools or analyzing logs. A simple command for readable output is: **df -h**

ubuntu@tryhackme:~\$ **df -h**

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	--G	12G	--G	17%	/
tmpfs	1.9G	0	1.9G	0%	/dev/shm
tmpfs	774M	1.2M	773M	1%	/run
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	387M	192K	387M	1%	/run/user/1000
tmpfs	387M	172K	387M	1%	/run/user/114

The **-h** means "human readable"; it shows sizes like 2G or 500M instead of long bytes-only numbers.

Breakdown of the Information

- **/dev/root** is the main disk of the system with --G total, 12G used, <REDACTED>G free, and is 17% full.
- **tmpfs** entries are temporary filesystems stored in RAM, not on the physical disk.
- **/dev/shm** is a shared memory area with 1.9G available and 0 used.
- **/run/user/114** is similar temporary storage for another system user, also 387M total and mostly empty.

Step 4: Read a System File

Linux stores configuration and informational files in the **/etc** directory. To practice navigating and reading files, head into **/etc** by running **cd /etc** and then list what's inside: **ls**

Use **cat** to read the os-release file that almost every Linux system contains: **cat os-release**

This file provides details about the Linux distribution that are often clearer than those provided by the **uname** command.

Q1) What is the username returned by the **whoami** command?

Answer: ubuntu

Q2) What is the kernel version shown by **uname -a**?

Answer: 6.14.0-1018-aws

Q3) How much free disk space does **df -h** report?

Answer: 58G

Q4) What is the message written inside day1_report.txt?

Answer: END-OF-DAY1

Command: **find -name day1_report.txt**

cat ./home/ubuntu/.logs/archive/day1_report.txt